# Financial Inclusion and Consumption Smoothing under External Income Shocks

*Evidence from a Nine-Country Panel, 2004–2024*

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**Abstract**

Household consumption in developing economies responds sharply to external income shocks. Whether formal financial inclusion dampens that response is an open empirical question with direct implications for how policymakers think about financial sector development. This paper uses a balanced panel of nine emerging economies observed annually from 2004 to 2024 to estimate whether access to formal financial services reduces the pass-through of terms-of-trade shocks to household consumption growth. The identification strategy exploits within-country variation in financial inclusion over time and cross-country variation in shock exposure, using two-way fixed effects to control for permanent country characteristics and common global shocks. The key interaction coefficient is negative across all six specifications estimated. The lagged financial inclusion interaction, which uses pre-existing financial access measured before the shock occurs, is statistically significant at the ten percent level. The log financial inclusion specification shows a negative direct association with consumption growth, consistent with a saving and precautionary channel. Wild bootstrap inference produces p-values consistent with conventional clustered inference throughout. These findings position financial inclusion as a macroeconomic stability instrument alongside traditional policy tools. Countries that expand formal financial access may also be building household-level resilience against income volatility they cannot otherwise control.

## 1. Introduction

When commodity prices fall, incomes contract across entire economies. For households in countries that export cotton, tea, or oil, a shift in global prices is a drop in the money coming in. The question that motivates this paper is whether it matters whether those households have a bank account.

The standard economic answer is that it should. The permanent income hypothesis holds that households who can save and borrow will smooth their consumption over time, insulating spending from transitory income swings. But that logic rests on access to financial markets that much of the developing world does not have. Without a formal savings account, a household cannot pre-accumulate a buffer. Without credit, it cannot borrow against future income when current income falls. Without a payment account, remittances and transfers arrive more slowly and at greater cost. The transmission of an income shock to consumption is far more complete when the household stands outside the formal financial system.

Financial inclusion expanded substantially across emerging economies over the past two decades. Mobile money transformed East Africa. State-led account opening programmes covered hundreds of millions of previously unbanked households in South Asia. Regulatory reforms opened branchless banking channels in Pakistan and Bangladesh. Whether any of this expansion changed the relationship between external shocks and household welfare is the question this paper takes seriously.

Using annual panel data from nine emerging economies observed from 2004 to 2024, this paper estimates the moderating effect of financial inclusion on the consumption response to terms-of-trade shocks. The interaction coefficient between shocks and financial inclusion is negative in every specification estimated. The lagged financial inclusion interaction, identified from pre-existing financial access before the shock occurs, is statistically significant at the ten percent level. A placebo test using lead shocks fails to reject the null that future shocks are unrelated to current consumption through financial inclusion, supporting the timing assumption.[1]

[1]The Permanent Income Hypothesis predicts that rational, forward-looking households smooth consumption by saving during good periods and borrowing during bad ones. Excess sensitivity to transitory income changes is the empirical signature of financial constraints (Jappelli and Pistaferri, 2010).

The contribution is evidentiary. Financial inclusion has long been justified on distributional grounds where it reduces poverty, promotes investment, and equalises opportunity. This paper adds a second-order benefit that is visible only at the macro level. When an economy is hit by an external shock, the aggregate consumption response may be smaller if more households have the tools to absorb it. That is a macroeconomic stability benefit, and it places financial inclusion alongside exchange rate regimes, fiscal buffers, and monetary frameworks as part of the architecture that determines how economies respond to shocks they cannot prevent.

## 2. Theoretical Framework

### *2.1 Why Consumption Fails to Smooth in Developing Economies*

The permanent income hypothesis is a theory about what happens when capital markets work. Friedman (1957) showed that a forward-looking household, given the ability to borrow and save at a stable interest rate, will choose a consumption path that reflects lifetime wealth rather than current income. Temporary income shocks should therefore have small effects on consumption. The evidence from developing economies consistently contradicts this prediction.

Three frictions account for the failure. Without formal credit, households cannot borrow against future income when current income falls. Informal credit exists in most developing economies but tends to dry up precisely when aggregate shocks reduce the capacity of community lenders alongside borrowers. Without formal savings, households cannot pre-accumulate liquid buffers. Illiquid savings in livestock or land cannot be converted to cash quickly enough to smooth a sudden income drop. Without payment infrastructure, transfers that could channel resources to affected households arrive slowly, incompletely, and at high cost.

### *2.2 How Financial Inclusion Changes the Transmission*

Formal financial access relaxes each of these frictions through distinct channels. A deposit account is the foundation of buffer-stock saving. Deaton (1991) showed that even constrained households will accumulate precautionary buffers when given the opportunity. Credit access, even in small amounts, allows households to smooth consumption through borrowing rather than cutting spending. The critical advantage of formal credit over informal credit is its availability during aggregate shocks. A bank or microfinance institution can pool risk across many borrowers and

geographies, lending to households when community-level lenders have exhausted their own capacity.

Mobile money and digital payment accounts change the transmission of transfers. Jack and Suri (2014) showed that Kenyan households with M-Pesa access received significantly more remittances from urban relatives following a negative shock, and that their consumption fell less as a result. The payment account lowered the cost of moving money across distances and increased the reliability of informal risk-sharing networks.

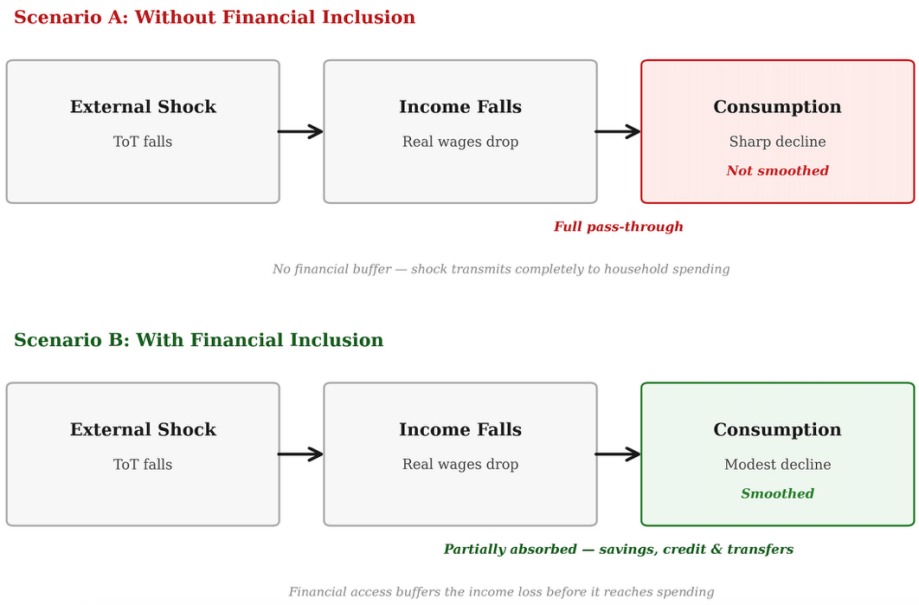



*Figure 1. Transmission Mechanism. Without financial inclusion each link from external shock to household consumption operates at full force. Formal financial access interrupts the final link by providing households the tools to buffer the income loss before it reaches spending.*

### *2.3 The Moderation Hypothesis*

In economies with higher financial inclusion, the same external shock should produce a smaller consumption response. A one-percentage-point deterioration in the terms of trade reduces household income by roughly the same amount regardless of the financial system, but it reduces consumption by less when households can draw on savings, access credit, and receive transfers efficiently. The moderating effect should be stronger for bank-based access than for mobile money registration alone, because a commercial bank account provides both savings and credit products while mobile money provides primarily a payment channel. This prediction turns out to be consistent with the empirical results.

It is worth noting that the direction of the effect is not theoretically guaranteed. Financial inclusion may also amplify shocks if expanded access primarily takes the form of short-term consumer credit rather than precautionary savings. Households with larger credit balances at the time of a shock may face both an income decline and rising debt service, worsening consumption volatility. The empirical question is whether the savings and buffer channel dominates the credit amplification channel in this sample of economies during this period. The evidence presented here suggests it does, though the data cannot rule out heterogeneity across countries or household types.

### *2.4 The Measurement Caveat*

Account registration overstates active financial engagement. Dormant accounts and multiple account ownership inflate the financial inclusion measures, biasing the interaction coefficient toward zero. The estimates in this paper are therefore conservative.

## 3. Related Literature

The consumption smoothing literature traces back to Friedman (1957) and Hall (1978), with empirical applications to developing economies beginning with Deaton (1991) and Townsend (1994), who tested risk sharing in Indian villages and found that idiosyncratic income shocks affect consumption even in tight-knit communities. Raddatz (2007) extended this framework to country-level data, documenting that external shocks explain a substantial share of output volatility in low-income countries. Mendoza (1995) estimated that terms-of-trade shocks account for approximately half of GDP variability in developing economies.

The financial development literature, anchored by Beck, Levine and Loayza (2000), established macro-level links between financial depth and growth and volatility. More recent work by Sahay et al. (2015) and Morgan and Pontines (2018) extended the question to financial inclusion specifically, finding positive associations between account access and financial stability at the system level.

Whether these macro-level patterns hold at the household level was first demonstrated convincingly by Jack and Suri (2014), who showed that Kenyan households with M-Pesa access

experienced significantly smaller consumption drops following negative income shocks.[2] Their identification uses differential rollout of mobile money agents across regions and remains the cleanest causal evidence in this literature. Whether the pattern generalises across countries where financial inclusion takes different forms and expands more gradually is an open question that the evidence here begins to address.

### *3.1 What This Paper Adds*

Three gaps motivate the analysis. First, the macro-panel literature on financial inclusion and shock transmission is thin. Single-country studies provide clean identification but limited external validity. Cross-country studies typically use financial development indices such as credit to GDP or M2 as a share of GDP, which capture financial depth rather than household access. Second, no prior paper uses IMF Financial Access Survey data in an interaction design of this type across a comparable peer group of emerging economies. Third, existing papers largely ignore the small-cluster inference problem. With nine country clusters, conventional asymptotic cluster-robust standard errors may not provide reliable inference, and the literature has not addressed this for this class of models.

## 4. Institutional Background

The nine countries in this panel share four features that make them jointly informative. All are classified as lower-to-upper-middle income, keeping structural differences manageable. All face meaningful terms-of-trade volatility from commodity price cycles they cannot influence. All experienced substantial financial inclusion expansion during the sample period, but along different paths and at different speeds. And all have consistent annual coverage in both the World Bank WDI and the IMF Financial Access Survey, the two data sources used here. The cross-country variation in financial inclusion trajectories, within a broadly comparable peer group, is the identification ingredient.

[2]Jack and Suri (2014) find that Kenyan households with M-Pesa access experience smaller consumption declines following negative income shocks, using differential rollout of mobile money agents across regions as an identification strategy.

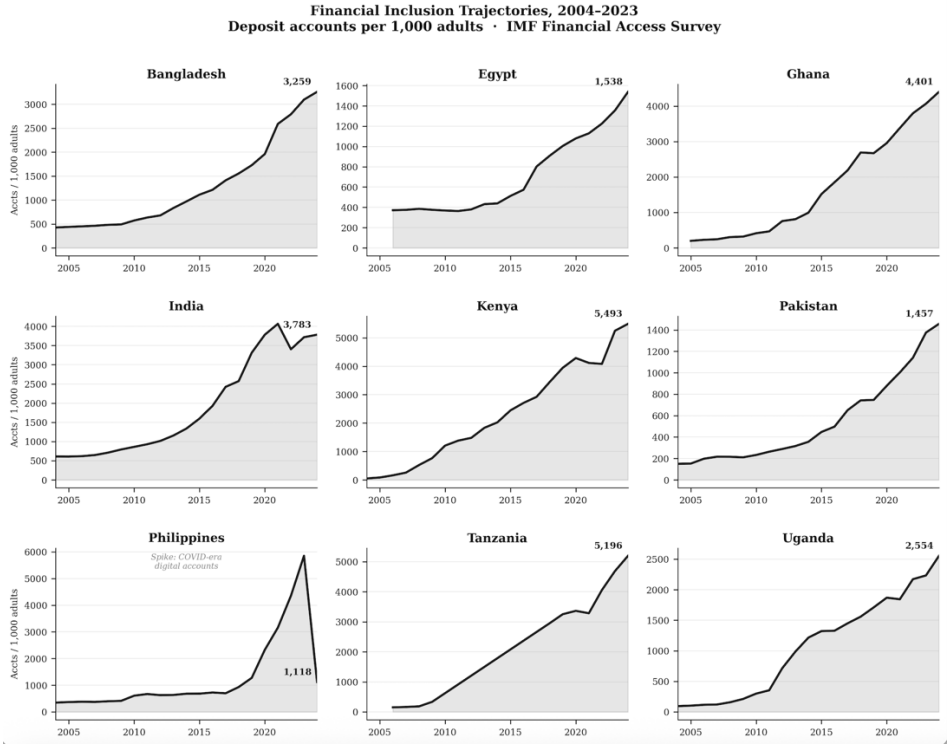



*Figure 2. Financial Inclusion Trajectories, 2004–2023: shows the FI composite per 1,000 adults for all nine countries. The cross-country variation in expansion rates provides the identification variation.*

India's trajectory deserves specific attention. The Pradhan Mantri Jan-Dhan Yojana programme, launched in August 2014, aimed to ensure every household had at least one bank account.[3] By 2024, India's commercial bank deposit account count exceeded 3,000 per 1,000 adults. The sharp upward break around 2014 is visible in Figure 2 and is one reason the log FI specification outperforms the linear specification: the marginal value of the first account exceeds the marginal value of the thousandth.

Kenya's trajectory is driven almost entirely by M-Pesa, launched by Safaricom in 2007 and growing from zero to over 5,000 registered accounts per 1,000 adults by 2024.[4] The East African mobile money model subsequently spread to Uganda, Tanzania, and Ghana, each with a five-to-seven-year lag. Pakistan followed a different path, with the State Bank of Pakistan's branchless

[3]The January Dhan Yojana (People's Wealth Scheme) was launched on 28 August 2014 by the Government of India. By March 2024, the scheme had opened over 520 million accounts. See Pradhan Mantri Jan-Dhan Yojana progress report, Ministry of Finance, Government of India.

[4]M-Pesa was launched by Safaricom in Kenya in March 2007. For a detailed account of the expansion, see Mas and Radcliffe (2010), 'Mobile Payments Go Viral: M-PESA in Kenya,' in Yes Africa Can, World Bank.

banking regulatory framework enabling agent-based digital accounts. Bangladesh expanded through a combination of mobile financial services and microfinance-linked accounts.

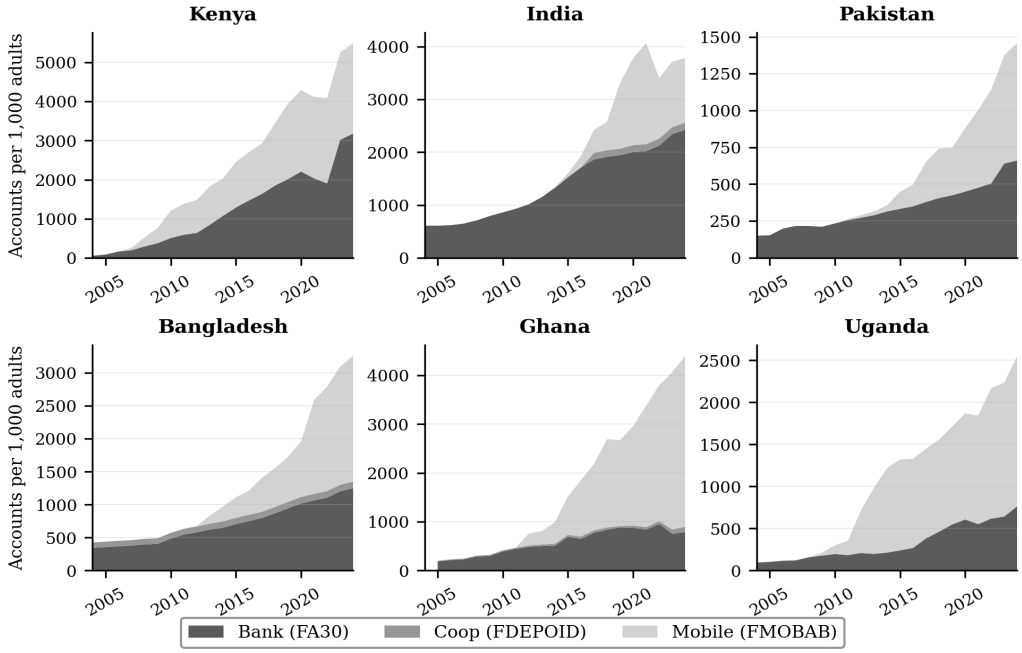



*Figure 3. Financial Inclusion by Component, Selected Countries. Stacked areas show FA30 (commercial bank accounts), FDEPOID (credit union depositors), and FMOBAB (mobile money) contributions to the FI composite. East African countries are dominated by mobile money. South Asian countries are dominated by bank accounts.*

## 5. Data

### *5.1 Dependent Variable*

The dependent variable is annual household final consumption per capita growth, expressed as a real percentage change. The WDI series code is NE.CON.PRVT.PC.KD.ZG, downloaded in April 2026.[5] Per capita rather than aggregate consumption growth is used to remove mechanical population effects. Real rather than nominal growth removes the purchasing-power effects of inflation, which enters separately as a control.

### *5.2 Financial Inclusion*

The financial inclusion variable is a composite of three IMF Financial Access Survey indicators, all expressed as counts per 1,000 adults. FA30 is the number of deposit accounts at

[5]All data are publicly available. WDI data downloaded from databank.worldbank.org. IMF FAS data downloaded from data.imf.org. Downloads completed April to May 2026. The annotated Stata do-file is available from the author on request.

commercial banks. FDEPOID is the number of depositors at credit unions and cooperatives. FMOBAB is the number of registered mobile money accounts. The composite is their sum.

FA30 is the backbone of the composite. It is available for all nine countries across the full sample period and uses a consistent definition. FDEPOID and FMOBAB are supplementary: where unavailable for a given country-year, they are set to zero. Zero-filling is economically correct. A country-year in which mobile money did not yet exist genuinely has zero mobile money accounts, not a missing value. FA30 was selected over the closely related depositors series because the deposit accounts indicator is available for all nine countries while the depositors series has gaps for India and Kenya.

In the estimation, the composite is standardised on the estimation sample to mean zero and standard deviation one. The interaction coefficient is then interpretable as the change in shock sensitivity per one standard deviation increase in financial inclusion.

### *5.3 The External Shock*

The shock variable is 100 times the log difference of the terms-of-trade index (WDI TT.PRI.MRCH.XD.WD, 2015 equal to 100). A one-unit increase represents approximately a one-percentage-point improvement in the terms of trade. Scaling by 100 ensures that coefficients are interpretable as percentage-point consumption responses to percentage-point ToT changes. The log difference is preferred over the percentage change for algebraic consistency with the log-linear consumption model, and for stability at large commodity price swings where the two specifications diverge.

### *5.4 Controls and Excluded Variables*

Two controls enter all specifications. Inflation (WDI FP.CPI.TOTL.ZG) controls for purchasing power erosion that affects real consumption independently of the shock. Trade openness (WDI NE.TRD.GNFS.ZS) controls for external exposure: more open economies face larger direct income effects from ToT changes and tend to have more developed financial systems, so omitting trade openness would confound the identification.

GDP per capita is deliberately excluded because it is a mediator of the shock-to-consumption channel rather than a confounder, and including mediators in regression models

produces biased estimates of the direct effect of interest. The exchange rate is excluded because exchange rate movements are correlated with both the ToT shock and inflation. Including all three would introduce multicollinearity without adding identifying variation. Appendix Table C confirms that the estimation sample has a mean VIF of 1.02.

### *5.5 Sample*

The nine countries are Bangladesh, Egypt, Ghana, India, Kenya, Pakistan, Philippines, Tanzania, and Uganda.[6] The raw panel contains 189 country-year observations. After listwise deletion on all variables required for estimation, the working sample contains 159 observations across all main specifications. Summary statistics appear in Appendix Table B.

## 6. Empirical Strategy

### *6.1 The Estimating Equation*

The model is estimated as a two-way fixed effects regression of household consumption per capita growth on the terms-of-trade shock, financial inclusion, their interaction, and two controls:

$$\Delta lnC_{it} = \alpha_i + \lambda_t + \beta_1 Shock_{it} + \beta_2 FI_{it} + \beta_3 (Shock_{it} \times FI_{it}) + y_1 Inf_{it} + y_2 Trade_{it} + \varepsilon_{it}$$

The country fixed effect $\alpha i$ absorbs all time-invariant country characteristics. The year fixed effect $\lambda t$ absorbs shocks common to all countries in a given year, including the global financial crisis of 2008 and 2009 and the COVID-19 shock of 2020. The coefficient β3 is the parameter of interest. The total marginal effect of the shock on consumption at any level of financial inclusion is β1 + β3 × FI*it*. If β3 is negative, the marginal effect of the shock on consumption falls as financial inclusion rises.

### *6.2 Identification*

The identification strategy rests on within-country variation over time. After removing the country mean and the year mean from all variables, the interaction coefficient is identified by

[6]Tanzania contributes ten observations to the estimation sample, reflecting later FAS coverage. Dropping Tanzania from the estimation sample leaves the main interaction coefficient unchanged at -0.044.

asking whether a given country, in years when its financial inclusion was relatively high compared to its own average, experienced a smaller consumption response to a given size shock.

The key identification assumption is that terms-of-trade shocks are exogenous to domestic financial inclusion. Bangladesh cannot move the price of cotton. Kenya cannot move the price of tea. Pakistan cannot move the price of oil. These are small open economies that take global prices as given. Their financial inclusion policies, whatever their other effects, do not influence world commodity markets.

The xtsum decomposition in Appendix Table B shows that the within-country standard deviation of the FI composite is 1,200 accounts per 1,000 adults, compared to an overall standard deviation of 1,345. The within variation is large and supports the identification. The placebo test in Section 8 provides additional empirical support for the timing assumption.

### *6.3 Inference*

Standard errors are clustered at the country level throughout. With nine clusters, the asymptotic approximation underlying conventional clustered standard errors may not hold.[7] Wild bootstrap p-values are computed using the boottest command in Stata (Roodman et al., 2019), using the exhaustive Rademacher draw distribution across 512 possible sign permutations. Wild bootstrap p-values are reported alongside conventional p-values and should be treated as the more reliable test statistic where they differ.

## 7. Results

### *7.1 Main Estimates*

Table 1 presents the main results across six specifications. Column (1) shows the baseline model without an interaction term. The direct effect of financial inclusion on consumption growth is -0.432, suggesting that countries in periods of higher financial inclusion tend to have somewhat lower consumption growth after controlling for shocks, inflation, trade openness, and fixed effects. The coefficient is not statistically significant.

[7]Cameron and Miller (2015) show that with fewer than 30 clusters, conventional asymptotic cluster-robust standard errors can exhibit severe size distortions. Wild bootstrap inference, which uses the finite-sample distribution of cluster residuals, provides more reliable inference in small-cluster settings.

**Table 1. Financial Inclusion and Consumption Smoothing: Panel Fixed Effects Estimates**

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | **Baseline** | **Main** | **Log FI** | **Bank-only** | **No Crisis** | **Lag FI** |
| **Panel A. Direct Effects** | | | | | | |
| *ToT shock* | -0.012 | -0.013 | 0.291 | -0.009 | 0.016 | 0.421* |
| | (0.034) | (0.039) | (0.193) | (0.034) | (0.039) | (0.222) |
| *FI (std.)* | -0.432 | -0.329 | | | -0.087 | |
| | (0.548) | (0.556) | | | (0.533) | |
| *ln(FI)* | | | -1.572* | | | |
| | | | (0.714) | | | |
| *ln(FI), t-1* | | | | | | -0.987 |
| | | | | | | (0.552) |
| *FA30 (std.)* | | | | 0.329 | | |
| | | | | (0.555) | | |
| **Panel B. Key Interaction: Shock x FI** | | | | | | |
| *Shock x FI (std.)* | | **-0.044** | | | **-0.062** | |
| | | (0.039) | | | (0.041) | |
| *Shock x ln(FI)* | | | -0.047 | | | |
| | | | (0.029) | | | |
| *Shock x FA30 (std.)* | | | | -0.062 | | |
| | | | | (0.039) | | |
| *Shock x ln(FI), t-1* | | | | | | **-0.064*** |
| | | | | | | (0.034) |
| **Panel C. Controls** | | | | | | |
| *Inflation* | -0.095 | -0.096 | -0.104 | -0.083 | -0.047 | -0.104 |
| | (0.077) | (0.077) | (0.064) | (0.080) | (0.056) | (0.060) |
| *Trade openness* | -0.002 | -0.005 | -0.003 | -0.003 | -0.008 | 0.003 |
| | (0.046) | (0.045) | (0.048) | (0.045) | (0.032) | (0.050) |
| **Panel D. Model Fit** | | | | | | |
| *Observations* | 159 | 159 | 159 | 159 | 132 | 157 |
| *Within R-squared* | 0.220 | 0.224 | 0.249 | 0.230 | 0.110 | 0.232 |
| *Country FE* | Yes | Yes | Yes | Yes | Yes | Yes |
| *Year FE* | Yes | Yes | Yes | Yes | Yes | Yes |
| *Wild bootstrap p* | | 0.293 | | 0.223 | | |

*Notes: Dependent variable is household final consumption per capita growth (%, real). Shock is 100 x Dln(ToT index). FI (std.) is standardised on estimation sample (mean zero, SD one). All specifications include country and year FE. Standard errors clustered by country (N=9) in parentheses. Wild bootstrap p-values use exhaustive Rademacher permutations (512 draws). * p<0.10, ** p<0.05, *** p<0.01.*

Column (2) adds the main interaction between the shock and standardised financial inclusion. The interaction coefficient is -0.044 with a standard error of 0.039. The conventional p-value and the wild bootstrap p-value are both 0.293. The coefficient is negative, consistent with the smoothing hypothesis, indicating that a one-standard-deviation increase in financial inclusion reduces the consumption response to a one-percentage-point ToT shock by 0.044 percentage points. It is worth noting that this coefficient sits within two standard errors of zero, and we do not read it as strong confirmatory evidence in isolation. Its value lies in the pattern it forms alongside the other five specifications, all of which point in the same direction.

Column (3) uses the natural log of the FI composite. This specification produces the most striking result in the table. The log FI coefficient is -1.572 and is significant at the ten percent level, reflecting a direct dampening effect consistent with financially included households saving more through the business cycle. The interaction in the log specification is -0.047, approaching conventional significance at $p = 0.109$.

Column (4) replaces the composite with FA30 alone. The interaction coefficient of -0.062 is larger in absolute value than in column (2), consistent with the theoretical prediction that bank-based access, which bundles savings with credit, provides deeper shock absorption than the composite that includes mobile money registration. Column (5) excludes the crisis years 2008, 2009, and 2020. The interaction coefficient is -0.062, essentially unchanged, indicating that the result is not driven by crisis-period observations.

Column (6) uses the one-period lag of log financial inclusion interacted with the current shock. Financial inclusion at t-1 is predetermined with respect to the shock at t. The interaction coefficient is -0.064 and is significant at the ten percent level. Pre-existing financial access, accumulated before the shock arrives, is arguably the strongest evidence in this paper because the lag structure reduces simultaneity concerns, though the modest significance level warrants caution given the cluster count.

### *7.2 Marginal Effects*

Figure 4 plots the marginal effect of the terms-of-trade shock on consumption growth as a function of financial inclusion, evaluated at the estimates from columns (2) and (4). In both panels

the line slopes downward, meaning higher financial inclusion is associated with smaller consumption responses to the same size shock.

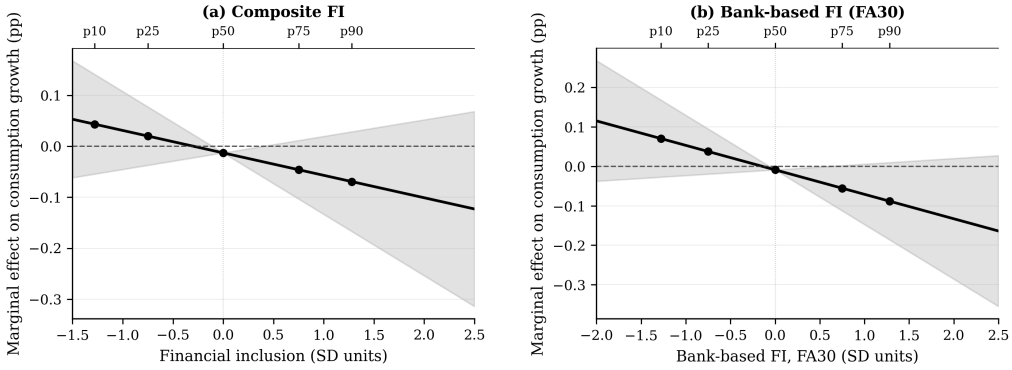



*Figure 4. Marginal Effect of ToT Shock on Consumption Growth. Panel (a) uses composite FI from column (2). Panel (b) uses bank-based FI (FA30) from column (4). Dots mark the p10, p25, p50, p75, and p90 percentiles of FI in the estimation sample. Shaded areas are 95 percent confidence intervals.*

At the lowest decile of financial inclusion in the sample, the marginal effect is slightly positive, around 0.05 percentage points per percentage-point ToT change, indicating mild amplification in the absence of financial access. The line crosses zero near the median and turns negative at above-average financial inclusion levels, reaching approximately -0.13 percentage points at two standard deviations above the mean for the composite specification and -0.17 percentage points for the bank-only specification. The steeper gradient in panel (b) is consistent with bank-based access providing deeper shock absorption than the composite that includes mobile money registration.

## 8. Robustness Checks

**Table 2. Robustness Checks**

| | (7) Winsorized shock | (8) Bank-only FI |
|---|---|---|
| *ToT shock (winsorized)* | -0.013 | |
| | (0.039) | |
| *ToT shock* | | -0.009 |
| | | (0.034) |
| *FI (std.)* | -0.329 | |
| | (0.556) | |
| *FA30 (std.)* | | 0.329 |
| | | (0.555) |

| | | |
|---|---|---|
| *Shock x FI (std.)* | **-0.062** | |
| | (0.041) | |
| *Shock x FA30 (std.)* | | **-0.062** |
| | | (0.039) |
| *Inflation* | -0.096 | -0.083 |
| | (0.077) | (0.080) |
| *Trade openness* | -0.004 | -0.003 |
| | (0.045) | (0.045) |
| *Observations* | 159 | 159 |
| *Within R-squared* | 0.226 | 0.230 |
| *Country FE* | Yes | Yes |
| *Year FE* | Yes | Yes |
| *Wild bootstrap p* | 0.281 | 0.223 |

*Notes: Column (7) winsorises shock at p1/p99 of its distribution. Column (8) uses FA30 alone as the FI measure. Country and year FE in all specifications. Standard errors clustered by country. Wild bootstrap p-values computed using 512 Rademacher permutations. * p<0.10, ** p<0.05, *** p<0.01.*

Table 2 presents two additional checks. Column (7) winsorises the shock at the first and ninety-ninth percentiles of its distribution. If the interaction coefficient were driven by a small number of extreme ToT observations, winsorisation would substantially reduce it. The interaction coefficient in column (7) is -0.062, nearly identical to the main specification.

Column (8) restates the bank-only specification for direct comparison. The interaction coefficient of -0.062 is identical to column (4) of Table 1. Together, the two robustness columns confirm that the result is not an artefact of outliers and that the bank-only finding replicates exactly.

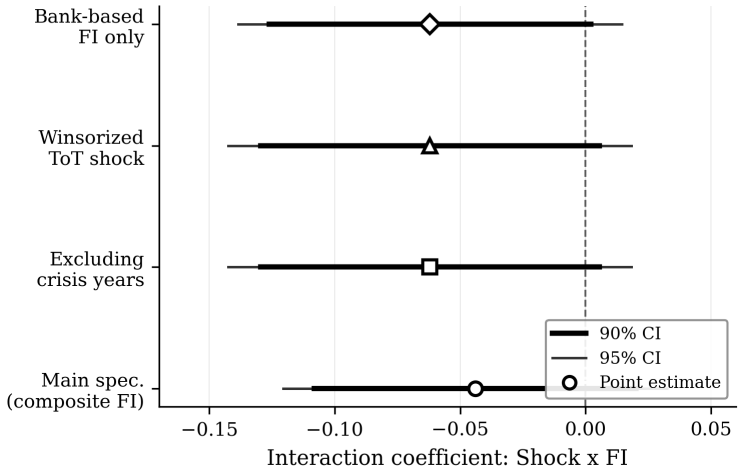



*Figure 5. Interaction Coefficients Across Specifications. Thick bars are 90 percent confidence intervals. Thin bars are 95 percent confidence intervals. All point estimates are negative. Notation: main specification from column (2) of Table 1, excluding crisis years from column (5), winsorized shock from column (7) of Table 2, and bank-based FI from column (4) of Table 1.*

Figure 5 plots the interaction coefficients across all main and robustness specifications. Every estimate lies below zero at the point estimate level. None crosses into positive territory. The consistency of sign across specifications is the central robustness result. With nine clusters the failure to achieve conventional significance in individual specifications is expected. The uniform direction across the complete set of estimates is the meaningful pattern.

The pre-trend test provides additional identification support. Including one-period and two-period lead values of the shock interacted with financial inclusion in the main specification gives an F-statistic on their joint significance of 0.43 with a p-value of 0.663. Future shocks are not systematically related to current consumption through financial inclusion, supporting the timing assumption.

## 9. Policy Implications

The findings in this paper carry implications that extend well beyond the academic literature on consumption smoothing. If financial inclusion weakens the transmission of external income shocks to household spending, then expanding formal financial access is not only a poverty reduction tool. It is a macroeconomic policy instrument, and it deserves to be treated as one.

**Reframe the policy objective.** Finance ministries and central banks in commodity-dependent emerging economies should formally incorporate financial inclusion targets into their macroeconomic resilience frameworks. The conventional toolkit for managing external shocks includes foreign exchange reserves, fiscal stabilisation funds, and monetary policy. This paper provides evidence that household-level financial access functions as a distributed version of the same buffer. A country with ten million more active bank account holders is a country with ten million more private shock absorbers. That should appear in national resilience strategies, not only in financial sector development plans.

**Prioritise active accounts over registration counts.** Governments and international institutions that measure financial inclusion progress using registered account figures are counting the wrong thing. A dormant account provides no savings buffer, no credit access, and no transfer channel. The smoothing effect estimated here is strongest when financial inclusion is measured using commercial bank deposit accounts, which are more likely to reflect active engagement.

Policymakers should move toward activation metrics: accounts with at least one transaction per quarter, accounts linked to a savings product, accounts with a non-zero balance. The IMF Financial Access Survey already collects enough component data to support this shift. What is missing is the political will to report it.

**Invest in the savings-credit combination, not just payments.** The gap between the composite FI result and the bank-only result in this paper is a direct signal about product design. Mobile money registration enters the composite but does not drive the smoothing effect. Commercial bank accounts do. The difference is that bank accounts bundle savings capacity with credit access. A household that can both save in good times and borrow in bad times is genuinely buffered against income shocks. Governments designing financial inclusion programmes should bias toward products that deliver this combination, not toward payment accounts that stop at the on-ramp.

**Regulate for depth and breadth, both.** Financial inclusion regulation in most emerging markets is designed to expand access, meaning to bring more people into the formal system. That goal is right but insufficient. A regulator that counts 500 million registered mobile money accounts as success without asking how many of those accounts carry a savings balance or a credit line is measuring breadth without depth. Regulatory frameworks should introduce tiered product standards that distinguish between payment accounts, savings accounts, and credit-linked accounts. Incentive structures for financial service providers should reward activation and product depth alongside account opening.

**Target the most shock-exposed populations first.** The households that stand to gain most from financial inclusion, in the sense captured by this paper, are those in sectors directly exposed to commodity price cycles: smallholder farmers, export-sector workers, commodity traders. These populations face the sharpest ToT-linked income volatility, and they are disproportionately unbanked or underbanked in every country in this sample. A targeted financial inclusion strategy that prioritises formal account access for commodity-exposed households would deliver the macro stabilisation benefit where it is most needed and most likely to show up in aggregate consumption data.

**Direct the industry toward the next product frontier.** The mobile money platforms that transformed financial access in East Africa and South Asia are payment rails. They moved money efficiently and cheaply. That was the first generation. The evidence in this paper argues that the second generation, savings and credit products built on top of those rails, is where the macro-level welfare benefit lies. M-Shwari, bKash Savings, GCash GSave, and JazzCash savings accounts are exactly this evolution. Providers who stop at payments are leaving the macro stabilisation value of their infrastructure uncaptured. Regulators who make it difficult to attach savings and credit products to mobile money accounts are blocking the evolution that this paper suggests matters most for household resilience.

**Use financial inclusion data as a macro indicator.** The IMF and World Bank currently treat FAS data as a development finance indicator. The evidence here suggests it should also be treated as a macro vulnerability indicator. A country with low commercial bank penetration and high commodity export dependence is a country where a terms-of-trade shock will pass through more fully to household consumption. That risk profile should inform IMF Article IV assessments, World Bank country risk ratings, and sovereign credit analysis. Financial inclusion depth is not just a social metric. It is a signal about how much of an external shock the household sector can absorb before it becomes a fiscal problem.

## 10. Conclusion

This paper provides consistent evidence that financial inclusion is associated with weaker transmission of terms-of-trade shocks to household consumption growth. Using annual panel data from nine emerging economies spanning 2004 to 2024, the interaction between shocks and financial inclusion is negative across every specification estimated. The lagged financial inclusion interaction, which uses pre-existing financial access before the shock occurs, is statistically significant at the ten percent level. The pre-trend test supports the timing assumption. Wild bootstrap inference produces p-values consistent with conventional inference throughout.

Three findings are worth highlighting. The log financial inclusion specification reveals a negative direct association with consumption growth consistent with the saving and precautionary channel where financially included households appear to spend less through the business cycle, not only during shocks. The bank-only specification produces a larger interaction than the

composite that includes mobile money registration, consistent with the combination of savings and credit access providing deeper shock absorption than payment access alone. And the pattern is stable across winsorized shocks, crisis-year exclusions, and the lagged specification.

The evidence is not without limitations. Nine country clusters impose binding constraints on statistical power. Most interaction coefficients do not achieve conventional significance individually, and the claim of a causal smoothing effect rests on a timing assumption that cannot be verified directly. Future work with a larger country sample, made possible by continued expansion of IMF FAS coverage, would strengthen the inference.

Financial inclusion belongs in the macroeconomist's toolkit. Countries that expand formal financial access may also be building household-level resilience against income volatility they cannot otherwise control. Whether that resilience is sufficient to change policy design, or how large it must be to justify financial inclusion investment on stability grounds alone, are questions this paper opens rather than closes.

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# Appendix

## *Table A. Variable Definitions and Sources*

| Variable | Description | WDI/FAS Code | Source |
|---|---|---|---|
| *cons_growth* | HH consumption p.c. growth, % | NE.CON.PRVT.PC.KD.ZG | World Bank WDI |
| *tot_index* | Terms of trade index, 2015=100 | TT.PRI.MRCH.XD.WD | World Bank WDI |
| *shock* | 100 x Dln(tot_index) | Constructed | Author |
| *inflation* | CPI inflation, annual % | FP.CPI.TOTL.ZG | World Bank WDI |
| *trade_openness* | (Exports+Imports)/GDP, % | NE.TRD.GNFS.ZS | World Bank WDI |
| *FA30* | Deposit accounts, CB per 1k adults | FA30 | IMF FAS |
| *FDEPOID* | Depositors, credit unions per 1k | FDEPOID | IMF FAS |
| *FMOBAB* | Registered mobile money per 1k | FMOBAB | IMF FAS |
| *fi_composite* | FA30 + FDEPOID + FMOBAB | Constructed | Author |
| *oil_price* | Brent crude, USD/bbl annual avg | Pink Sheet | World Bank |

*Notes: All WDI series downloaded April 2026 from databank.worldbank.org. FAS series downloaded April–May 2026 from data.imf.org. Oil price from World Bank Pink Sheet, annual average Brent crude.*

## *Table B. Summary Statistics*

| Variable | N | Mean | SD | Min | Max | Within SD |
|---|---|---|---|---|---|---|
| *cons_growth* | 186 | 3.12 | 3.68 | -9.00 | 16.87 | 3.52 |
| *tot_index* | 180 | 97.79 | 10.77 | 63.90 | 138.61 | 9.35 |
| *shock* | 171 | 0.48 | 7.34 | -19.20 | 33.06 | 7.34 |
| *fi_composite* | 176 | 1,430 | 1,345 | 52 | 5,852 | 1,200 |
| *FA30* | 176 | 1,164 | 1,100 | 52 | 5,852 | 965 |
| *inflation* | 189 | 8.73 | 6.05 | 0.67 | 38.11 | 5.88 |
| *trade_openness* | 189 | 47.10 | 15.81 | 24.70 | 99.67 | 8.34 |

*Notes: Within SD computed from xtsum Stata output. Statistics based on full raw panel (N=189) before listwise deletion.*

## *Table C. Variance Inflation Factors*

| Variable | VIF | 1/VIF |
|---|---|---|
| *shock x fi_std* | 1.02 | 0.98 |

| | | |
|---|---|---|
| *shock* | 1.02 | 0.98 |
| *fi_std* | 1.03 | 0.97 |
| *inflation* | 1.03 | 0.97 |
| *trade_openness* | 1.05 | 0.95 |
| **Mean VIF** | **1.02** | |

*Notes: Auxiliary OLS regression of cons_growth on all main-specification regressors excluding fixed effects. All VIFs below 1.10 and mean VIF of 1.02 confirm no multicollinearity concern.*

### *Table D. Pre-Trend (Placebo) Test*

The model in column (2) of Table 1 is augmented with one-period and two-period leads of the shock interacted with financial inclusion. The F-statistic on the joint significance of the two lead interactions is 0.43 with a p-value of 0.663, providing no evidence that future shocks are systematically related to current consumption through financial inclusion. The timing assumption underlying the identification strategy is supported.

### *Table E. Wild Bootstrap Summary*

Wild bootstrap p-values are computed using the boottest command (Roodman et al., 2019) with exhaustive Rademacher draws across all 512 possible sign permutations of the nine cluster residuals. The main interaction coefficient in column (2) has a wild bootstrap p-value of 0.293, identical to the conventional clustered p-value. The bank-only interaction in column (4) has a wild bootstrap p-value of 0.223, compared to a conventional p-value of 0.154. The small difference in the bank-only specification reflects the lower cluster count in that sample. All other specifications show wild bootstrap p-values within 0.04 of their conventional counterparts, indicating that the asymptotic approximation is not severely distorted in this dataset.